Rishitha Rajapu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.35
Tirumala junior college	Class XII (Board of intermediate education, Andhra Pradesh)	2023	97.40%
Vivekananda High School	Class X (Andhra Pradesh Board of Secondary Education)	2021	100.00%

POSITIONS OF RESPONSIBILITY

Worked as coordinator in the Web Dev team of EML club Team 2025-2026

SKILLS

Programming Languages: Python, R, C, MySQL

Python Frameworks: Tensorflow, Pytorch, Langchain, Langgraph, MCP, Fast MCP, Django

Python Libraries: Numpy, Pandas, Matplotlib, OpenCV, OCR, Nltk, Sklearn

RELEVANT COURSES

Computer Science: Introduction to programming, DBMS 1, DBMS 2, Compilers 1, Computer Architecture, DSA, OS-1, Algorithms

Artificial Intelligence: Foundations of Machine Learning, Deep Learning, Reinforcement learning, Computer vision, Natural language processing, Robotics, Convex Optimization, Optimization-2, Artificial Intelligence, programming for AI

Mathematics: Discrete mathematics, Calculus-1, Calculus-2, Vector Calculus, Series of Functions, Probability and random variables, Differential equations, Matrix theory, Intro to Metric spaces, Applied Statistics, Bayesian statistics

WORK EXPERIENCE

Risk Analyst Intern | American Express | 8-week Summer Internship:

- Worked with the **Research & Algorithms team** to research and design **loss functions and model modifications** for improving fraud-risk model performance.
- Analyzed and experimented on a **40M+ transaction dataset** containing large-scale fraud transaction data to identify opportunities for improving existing risk models.
- Developed **two novel solutions** to enhance the existing fraud detection framework; both solutions **outperformed their benchmark approach** on key evaluation metrics.
- Achieved an estimated **$3.5M/month in savings**, demonstrating significant business impact through improved fraud-risk modeling.

PROJECTS

ML based Manipulation detection:

- Developed a **machine-learning based manipulation detection system** using the MentalManip dataset to classify

user utterances as manipulative or non-manipulative.
- Trained and evaluated several **heterogeneous classifiers** including Logistic Regression, Random Forest, SVM, KNN, Decision Tree, Gradient Boosting, AdaBoost, Extra Trees, Naive Bayes, and MLP to capture diverse decision boundaries.
- Built an **ensemble soft-voting framework** by averaging class-probability scores with a 0.5 decision threshold, using the highest-probability class when no classifier crossed the threshold.
- Integrated manipulation-type predictions with a **large language model (LLM)** by dynamically injecting safety instructions into prompts, enabling responses that remained **safe, polite, and robust** against manipulative user inputs.

Bayesian Genomic Modeling for Phenotype Prediction:

- Developed a **Bayesian statistical method for estimating polygenicity** and characterizing the genetic architecture underlying complex phenotypes from high-dimensional genomic data.
- Designed computational procedures for **Bayesian inference and parameter estimation**, accounting for uncertainty in genomic effect distributions.
- Developing an **R package** to make the proposed methodology accessible to researchers through a reproducible and user-friendly implementation.

Dataset distillation:

- Developed a **computationally efficient dataset condensation pipeline** on CIFAR-10 to generate compact synthetic datasets while reducing the resource overhead of Improved Distribution Matching (IDM).
- Proposed **exponential feature matching** to capture higher-order feature moments beyond conventional mean-based distribution matching, optimizing synthetic images through iterative feature alignment.
- Introduced **model weight perturbation** during synthetic image optimization as a lightweight alternative to maintaining memory-intensive model queues, improving robustness against overfitting to a single pretrained model.
- Replaced one-hot supervision with **soft-label training** using extracted synthetic feature representations and evaluated the condensed datasets across ConvNet, AlexNet, and ResNet architectures, achieving **45.2% accuracy at 10 images/class**.

Vehicle Rental System(Team Project)**:**

- A vehicle rental platform using MySQL and Django frameworks to manage customer bookings
- Ensured date-base availability so that customers can book available vehicles, view and cancel their orders and will be notified in case of late returns
- Built an admin dashboard to add/remove vehicles, view cancellations and manage dues